Introduction to Hematology

Blood types, Hemostasis and Hematological parameters

Integrated Biomedical and Clinical Sciences OMFS 512

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CALIFORNIA NORTHSTATE UNIVERSITY COLLEGE of DENTAL MEDICINE

Mission: To advance the science and art of health through education, service, scholarship and social accountability

Learning Objectives

1. Describe hematopoiesis and explain the different cell linage
2. Explain the blood components.
3. Describe oxygen-carbon dioxide transport mechanisms.
4. Explain the Hemoglobin metabolism and compare different types of anemia.
5. Describe molecular bases of different blood types.
6. Describe the clotting mechanism.
7. Compare and contrast different hematological parameter.

Outline

- Hematopoiesis
- Blood components
- Heme metabolism
- Transport of oxygen and carbon dioxide
- Heme and globin disorders
- Blood types and transfusion
- Clotting mechanism
- Hematological parameters

Functions of Blood

- Respiration
- Nutrition: transport of absorbed nutrients.
- Excretion through the transport of metabolic waste to various organs for removal.
- Regulation of acid-base balance, body temperature and water balance.
- Defense through WBCs and circulating antibodies.
- Transport of hormones and metabolites.
- Coagulation.

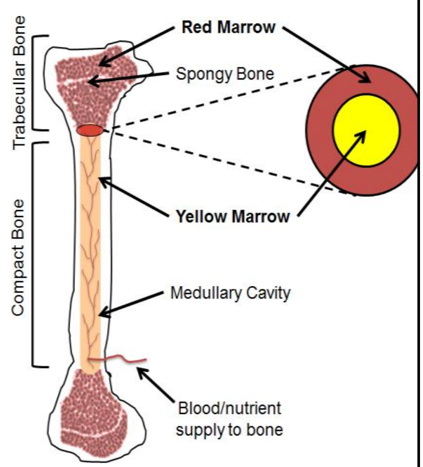
Hemopoiesis

(aka, Hematopoiesis)

Hemopoiesis refers to the development of blood cells. Can be:

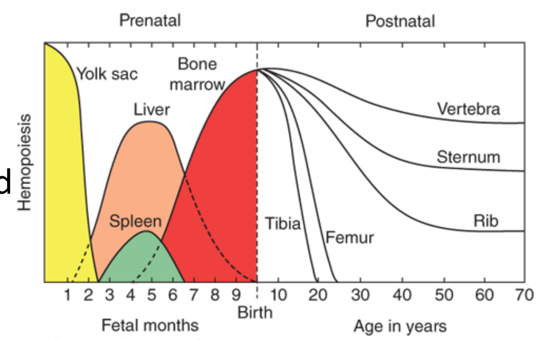
1- Medullary in bone marrow. The red marrow (containing the haemopoietic tissue). Each day, the marrow normally produces around billions of blood cells. Hematopoiesis continues throughout life but declines with aging.

2- Extramedullary: outside the bone marrow (spleen, liver and lymph nodes) during embryonic development.



(PDF) The Importance of Physiologically Inspired Physicochemical Parameters on Hematopoietic Stem-Cell Maintenance and Lineage Specific Differentiation in Ex Vivo Cultures (researchgate.net)

- During embryonic development, hematopoiesis starts in the yolk sac mesoderm. At this stage, limited to the production of monocytes, platelets and red blood cells.
- Hematopoietic stem cells (HSCs) arise around W4 of gestation, reside in the liver &start hematopoiesis by W6 throughout second trimester (liver is main site of hematopoiesis during fetal development).
- Bone marrow becomes the major hemopoietic organ during the last 2 months of gestation.



Source: Anthony L. Mescher: Junqueira's Basic Histology Text and Atlas, 16e

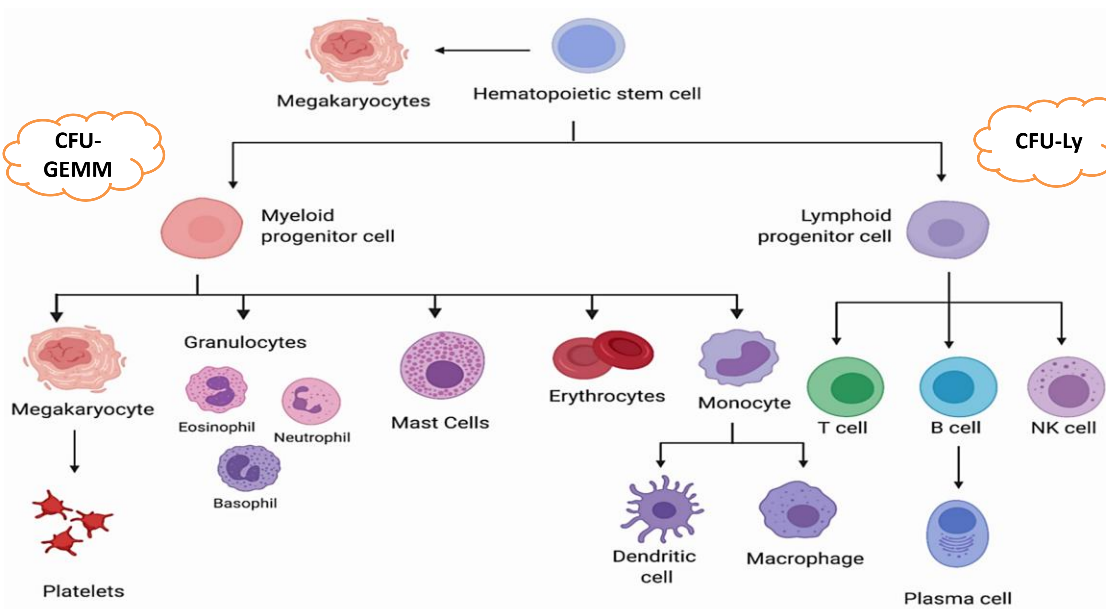
Hematopoietic stem cell

HSCs maintain hematopoiesis.

Multipotent cells capable of self-renewal and can remain quiescent for long periods of time. HSCs give rise to two major cell lineages of progenitor cells:

1- The myeloid lineage generates granulocytes, erythrocytes, monocytes and megakaryocytes (develop to platelets). Cells of myeloid lineage originate and develop in the bone marrow.

2-The lymphoid lineage forms lymphopoietic cells (T and B-cells as well as natural killer cells). Cells of the lymphoid lineage originate in the bone marrow, but they develop and mature in lymphoid organs “spleen and lymph nodes”



Raza Y, Salman H, Luberto C. Sphingolipids in Hematopoiesis: Exploring Their Role in Lineage Commitment. *Cells*. 2021; 10(10):2507. https://doi.org/10.3390/cells10102507

Regulation of Hemopoiesis

Cytokines and growth factors control mitotic rate, enhance HSCs survival and control their differentiation:

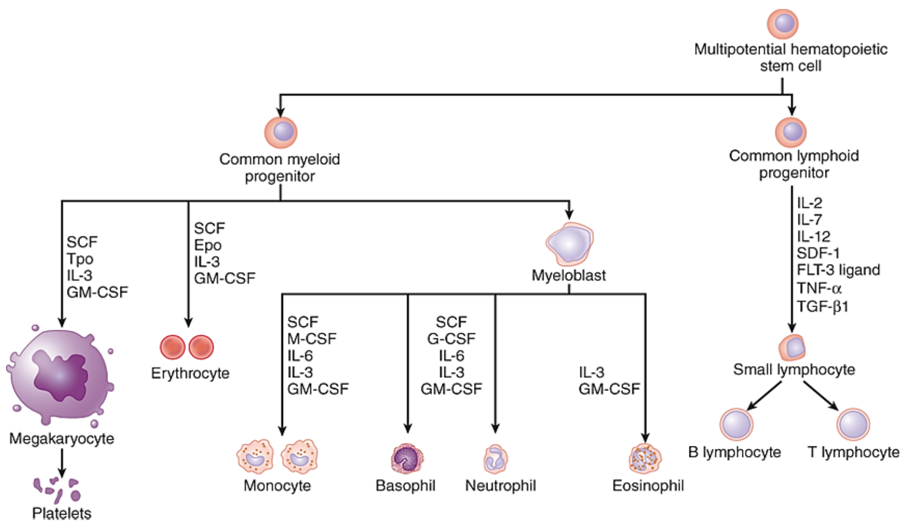
1- Stem cell factor, a glycoprotein produced by the bone marrow. Supports the production of all hemopoietic progenitor cells.

2- Erythropoietin, glycoprotein mainly produced by the kidneys. Promotes the production of erythrocytes.

3- Thrombopoietin, glycoprotein produced by the liver (main site) and kidney. Stimulates production and differentiation of megakaryocytes & platelets production.

4- Interleukin 1, 3 and 6 support early hematopoiesis in favor of myeloid progenitors.

5-Interleukin 2 and 7 aid in favor of lymphoid progenitors (T, B- and NK cells).



Source: Jon C. Aster. H. Franklin Bunn:

Hematopoiesis is tightly controlled by cytokines and growth factors. Uncontrolled hematopoiesis may lead to the development of *hematologic malignancies.*

[Hematopoiesis | Pathophysiology of Blood Disorders, 2e | AccessMedicine | McGraw Hill Medical (cnsu.edu)]

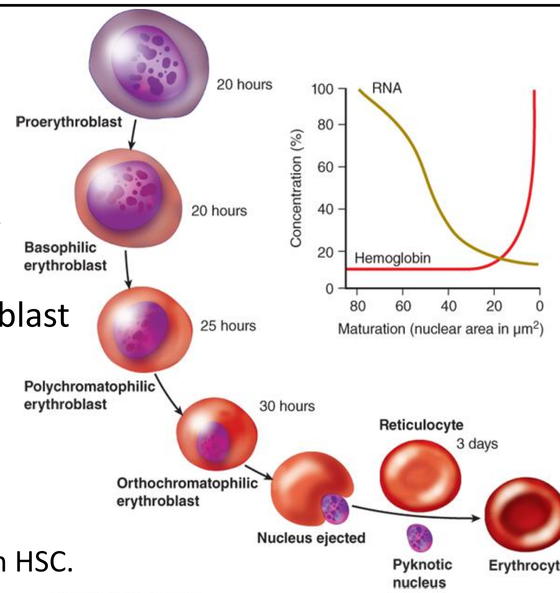
For illustrative purpose

Maturation of blood components

- Erythropoiesis describes the development of red blood cells from a proerythroblast.
- Granulopoiesis describes the development of granulocytes develop from myeloblasts.
- Monopoiesis describes the development of monocytes from monoblast cells. Next, they differentiate into macrophages, microglia or dendritic cells.

Erythropoiesis

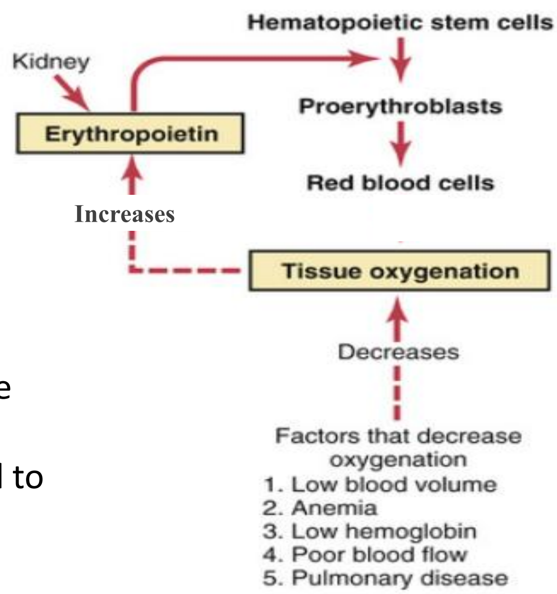
- Development of mature erythrocytes takes around 1 week and 3-4 cell divisions.
- Starts with the development of proerythroblast from a HSCs.
- Controlled by several genes, requires EPO, iron, vitamin B12, folic acid.
- Erythropoietin (EPO):
  - Stimulates production of proerythroblasts from HSC.
  - Speeds up HSCs mitotic rate and accelerate maturation.



Source: Anthony L. Mescher: Junqueira's Basic Histology Text and Atlas, 16e

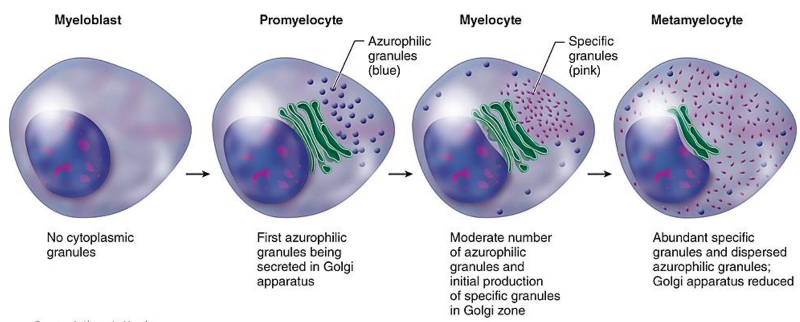
Decreased oxygen transport to the kidneys, stimulate erythropoietin secretion and increase the rate of RBC production (erythropoiesis) and inhibit eryptosis. [apoptosis pattern in mature RBCs].

Vitamin B12 (cobalamin) & folate are imp for DNA and RNA synthesis. Deficiency of either vitamin can lead to macrocytic anemia.



6- Decreased oxygen content (mountains)

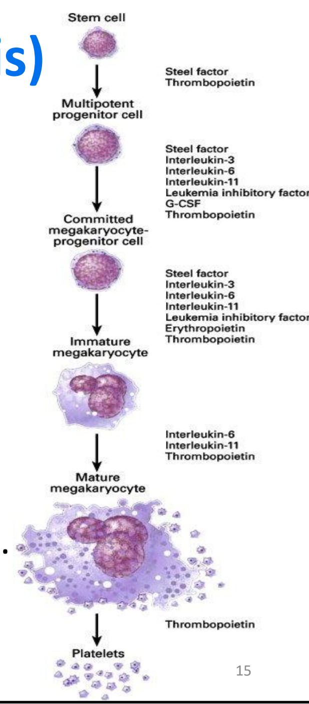
Granulopoiesis



Granulocytes develop from myeloblasts. Azurophilic granules containing lysosomal hydrolase enzymes are produced from the rough endoplasmic reticulum and secreted by Golgi apparatus. Granulocytes are imp. in innate immune response.

Megakaryocytopoiesis (thrombopoiesis)

- Megakaryoblast matures into a megakaryocyte. Mediated by thrombopoietin & interleukins (3, 6 and 11).
- Thrombopoietin induce the final maturation and regulates its production. Platelets (aka, thrombocytes) develop by fragmentation from mature megakaryocytes.
- Vitamin B12 and folic acid deficiency may lead to thrombocytopenia due to ineffective megakaryopoiesis. One megakaryocyte forms 4000 platelets.

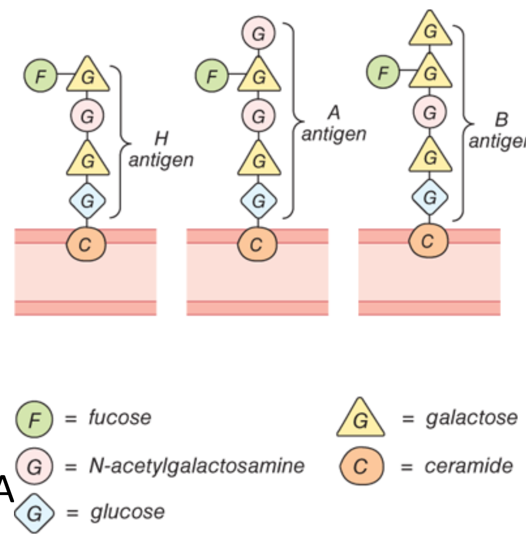


Thrombopoietin | NEJM

Blood types and Transfusion

Antigens of ABO System

- RBCs membranes contain several blood group antigens.
- *ABO* gene is located on chromosome 9. Encodes specific glycosyltransferases that synthesize RBCs antigens.
- An *H* gene forms the H antigen. H antigen acts as a precursor for antigen A or B.



Source: K.E. Barrett, S.M. Barman, H.L. Brooks, Jason X.J. Yuan: Ganong's Review of Medical Physiology, Twenty-Sixth Edition

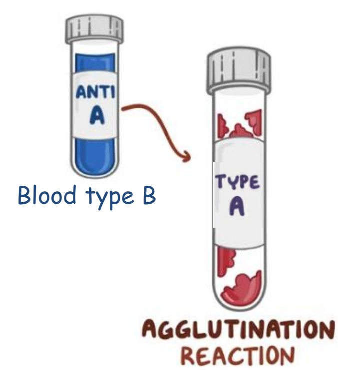
- Based on the antigen subtype:
  - Blood type A, has an A antigen (AA or AO).
  - Blood type B has a B antigen (BB or BO).
  - Blood type AB has both antigens (AB)
  - Blood type O has neither A or B antigens (just H antigens, OO).

Rare subtype: “Bombay phenotype” that lacks the H antigen. Can receive blood from other Bombay donors.

Anti-A and Anti-B antibodies

- These antibodies develop spontaneously during first six months of life. Antibodies are present in the blood plasma.

  - Blood type A, has an anti-B antibody.
  - Blood type B has an anti-A antibody.
  - Blood type AB **has neither** anti-A or anti-B.
  - Blood type O has both anti-A or anti-B.



Inheritance

A and B antigens are inherited in a dominant Mendelian pattern. Each parent passes an allele to offspring.

DAD	MOM		
	A	B	O
A	AA	AB	AO
B	--?	--?	--?
O	--?	--?	OO

offspring

RH factor

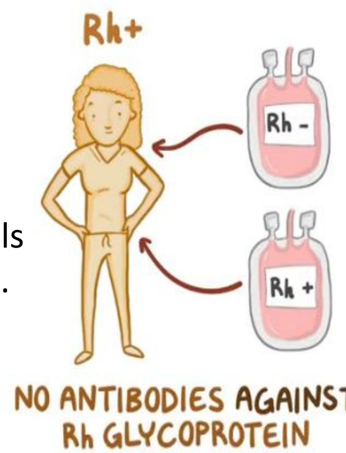
Rh antigens can be categorized into:

a) Rh positive: carries the D antigen (RhD) as well as C/E antigens (RhCE).

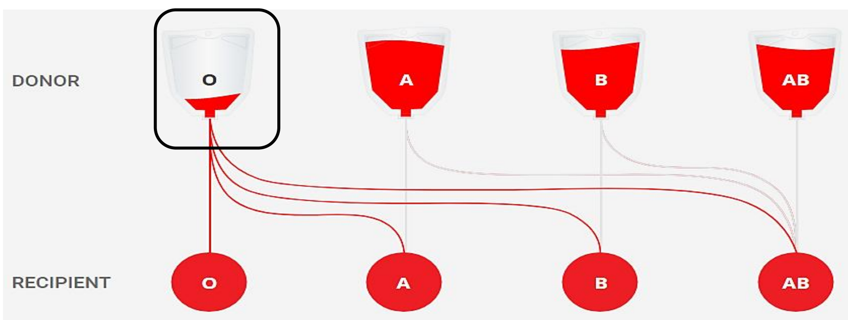
b) Rh negative: carries only RhCE antigen.

Rh is inherited independent from the ABO genes. Individuals are classified as D+ (or Rh-positive), or D– (or Rh-negative). Rh-positive phenotype is most prevalent (around 85 %).

- Rh^- can receive blood from Rh^- only.
- Rh^+ can receive from both.



Transfusion Compatibility



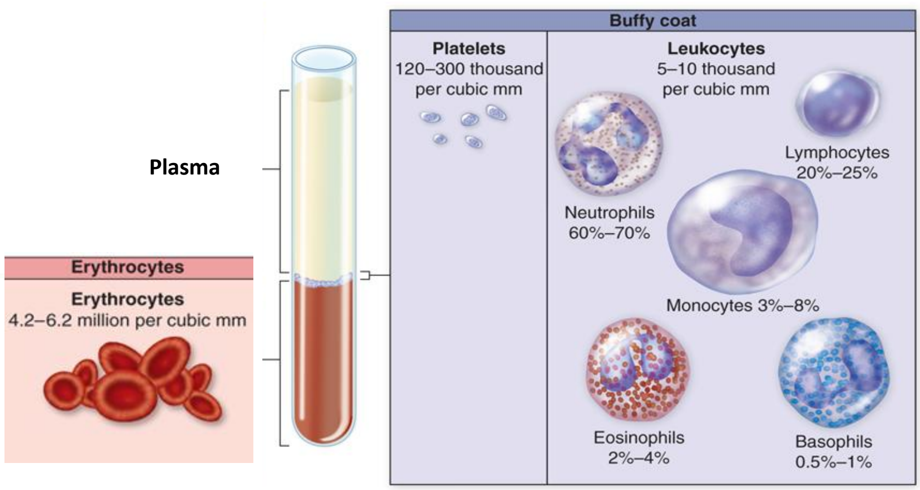
Coombs test *can detect the presence of antibodies against circulating RBCs. The presence of turbidity or cell aggregates indicate a positive agglutination reaction and hence the incompatibility of that blood type.*

Type "O" is a universal donor. "AB" is a universal recipient.

Blood Types Explained - A, B, AB and O | Red Cross Blood Services

Blood components

Blood is a fluid connective tissue composed of RBCs; WBCs & platelets suspended in plasma.



Plasma

- Make up 50 % of blood volume.
- Composed of water, electrolytes and plasma proteins.
- Plasma proteins are mainly synthesized in the liver. Maintain the oncotic pressure (blood osmotic pressure).
- A decrease of plasma proteins can lead to fluid accumulation in the extravascular tissues or edema.

Plasma		
Water 92% by weight	**Proteins** 7% by weight	**Other solutes** 1% by weight
	Albumins 58% Globulins 37% Fibrinogen 4% Regulatory proteins 1%	Electrolytes Nutrients Respiratory gases Waste products

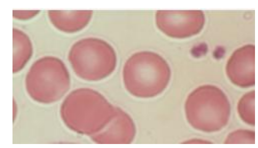
Plasma proteins

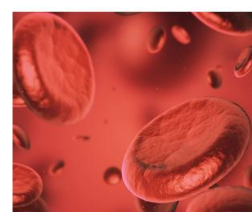
Include albumin, immunoglobulin, fibrinogen, transferrin and most components of complement and clotting cascades.

- Albumin: most abundant plasma protein. Important for maintaining oncotic pressure and as a carrier for numerous ligands & drugs.
  - Hypoalbuminemia can occur in liver failure. Decreases oncotic pressure leading to the leakage of fluid from blood into the tissues, resulting in edema.
- Fibrinogen is essential in blood clotting by forming fibrin fibers.
- Transferrin is a glycoprotein important for iron transport.

Red Blood cells

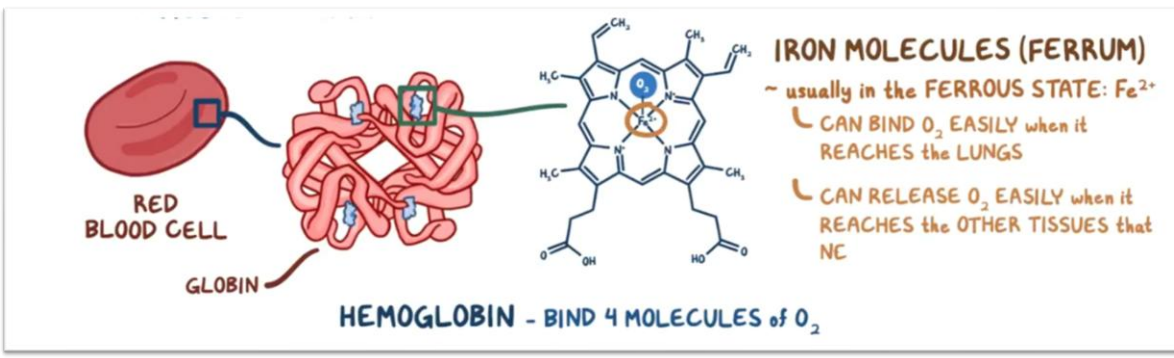
- Erythrocytes are biconcave cells, carry hemoglobin and perform oxygen transport in the circulation.
- Average survival time of 120 days in circulation. Average normal red blood cell count is 5.4 million/µL in males and 4.8 million/µL in females.
- RBCs has no nucleus and no mitochondria.
- Number, size of RBCs, as well as the amount and quality of carried hemoglobin are basic determinates for blood health.





Abnormalities associated with Red Blood cells

- Anemia is a common pathological condition attributed to reduced or abnormal hemoglobin, and/or reduced RBCs number. Metabolic enzymes deficiencies can lead to hemolytic anemia
- Erythrocytosis and polycythemia ="too many RBCs".
  - Erythrocytosis: Increased RBCs count in response to hypoxia or increased levels of erythropoietin.
  - Polycythemia Increased number of RBCs. Can be attributed to malignancy of RBCs that causes the over production of RBCs.



Hemoglobin is a metalloprotein crucial for oxygen transport.
Tetramer made up of 4 polypeptide globin chains and each globin contains a heme moiety.
The heme molecule is formed from a protoporphyrin ring and a central ferrous iron (ferroprotoporphyrin).

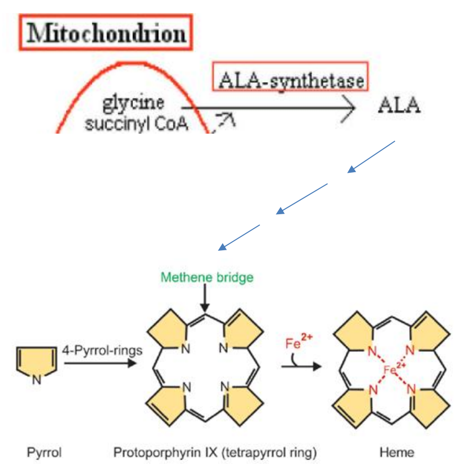
Hemoglobin synthesis

Globin Biosynthesis

- The globin chain is synthesized by ribosomes in the RBCs cytosol from alpha and beta globin chains.
- The α-globin genes are on chromosome 16 and β-globin genes are on chromosome 11.
- Heme, the porphyrin ring with an iron, associates with globin.

Heme Biosynthesis

- Occurs in mitochondria and cytosol.
- Starts from glycine in a multistep pathway. Final step: protoporphyrin IX is combined with iron (ferrous state) yielding heme which combines with globin to yield a subunit of hemoglobin.
- Heme can regulate its own synthesis via a negative feedback mechanism by inhibiting the mitochondrial aminolaevulinic acid (ALA) synthase.

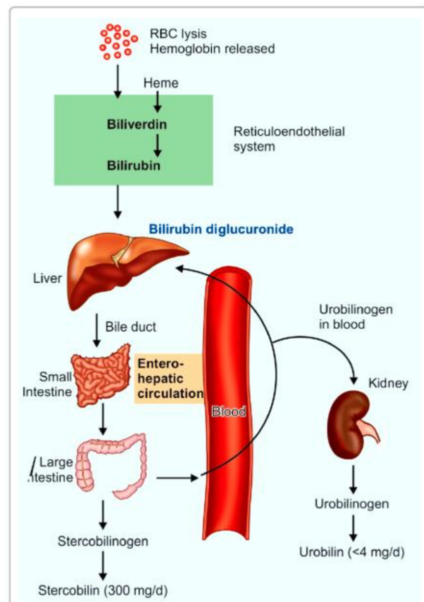


Heme biosynthesis pathway. ALA= d-aminolevulinic acid; ALA synthase is a rate limiting enzyme

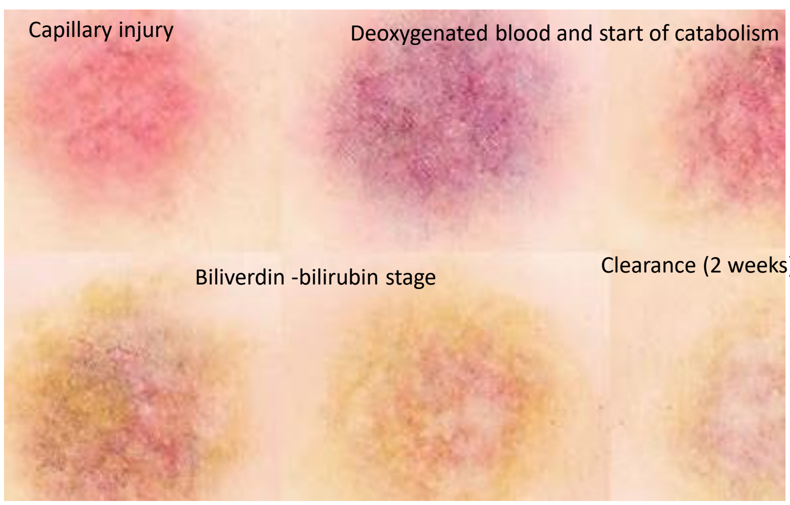
Fig. for illustration

Heme Catabolism

- Following RBCs lysis, released hemoglobin is immediately phagocytized by macrophages (mainly, liver, spleen and kidney).
- Globin join the amino acid pool. Heme is degraded by heme oxygenase system. Porphyrin is converted into biliverdin then to bilirubin.
- Liver uptakes the water insoluble bilirubin, convert to water-soluble diglucuronide (bile excreted).
- In the intestine, bilirubin diglucuronide is converted to urobilinogen (kidney excreted) and stercobilinogen (large intestine).



Sequence of Heme Catabolism in Contusions



Iron Metabolism

- Iron is absorbed from the gut or recycled from hemoglobin.
- Free iron is toxic. Stored in cells bound to ferritin protein and circulates in plasma bound to transferrin protein.
- Amount of absorbed iron is regulated according to body needs (liver & hepcidin levels).
- Liver synthesizes hepcidin (peptide hormone) to inhibit iron transport from the intestine wall to the blood.
- One milliliter of packed erythrocytes contains approximately 1 mg of iron.

In the body, iron is distributed between a functional (tissue) and storage pool.

- Tissue pool "majority of iron pool"

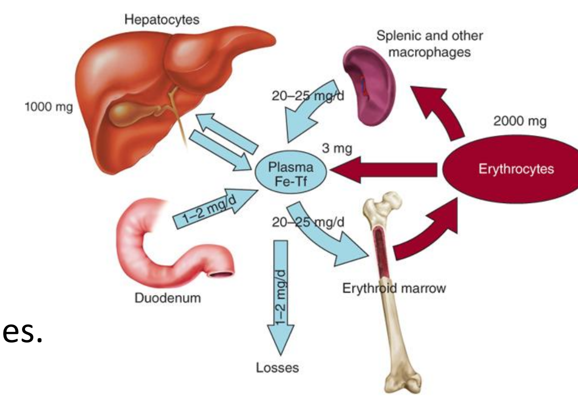
➢ Developing and mature erythrocytes as hemoglobin (around 65 % of iron pool).

➢ Skeletal muscles as myoglobin

➢ Cells in the form of iron-containing enzymes.

- Storage pool

Mainly in the liver, spleen and bone marrow.



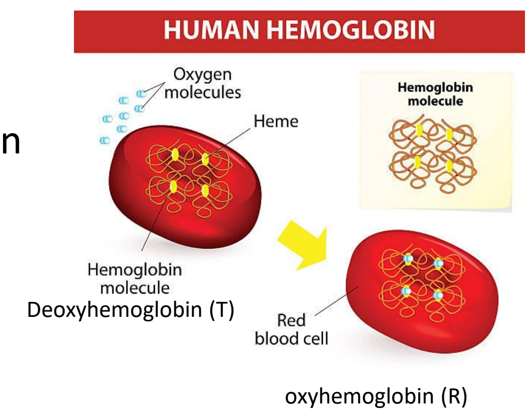
Source: Kenneth Kaushansky, Josef T. Prchal, Linda J. Burns, Marshall A. Lichtman, Marcel Levi, David C. Linch: Williams Hematology, 10e

[Iron Metabolism | Williams Hematology, 10e | AccessMedicine | McGraw Hill Medical (cnsu.edu)]

Functions of hemoglobin

1- Homeostatic function: Facilitate oxygen/carbon dioxide exchange. Hemoglobin cycle between an oxyhemoglobin (R) and deoxyhemoglobin (T) forms.

2- Enzymatic function: act as nitrate reductase to release nitric oxide (NO) from nitrite & promote vasodilation (distinct from eNOS mechanism).



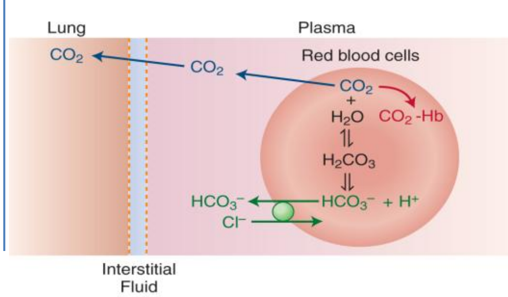
 Adapted from https://www.istockphoto.com/illustrations/hemoglobin

Homeostatic function:

A) Carbon dioxide Transport

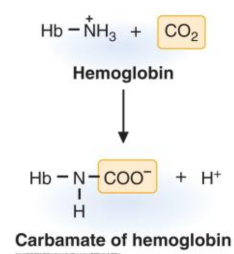
Around 10 % of carbon dioxide is dissolved in plasma

Majority is transported in the form of bicarbonate (60 %)



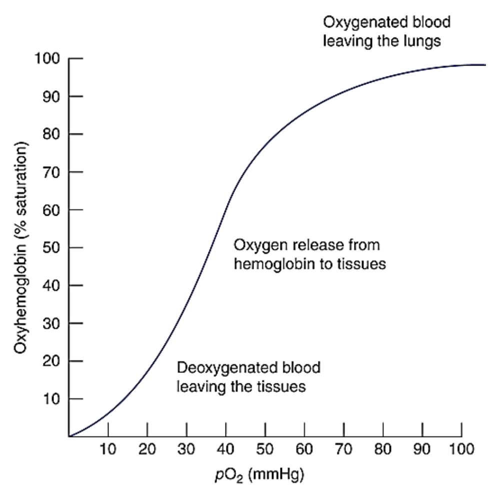
$CO_2 + H_2O \rightarrow H^+ + HCO3^-$ (bicarbonate)

Around 30 % binds to Hb as carbaminohemoglobin (carbamate of hemoglobin; stabilizes the deoxy Hb)



B) Oxygen Transport

- Hgb exist into states: tense (T) or relaxed (R), each with different affinity to oxygen. The “R” state has 100-fold higher affinity for oxygen.
- In the lungs, binding of the first O_2 to the heme induce confirmational change and increases hemoglobin affinity to bind more oxygen = transforming Hb to the “relaxed or high affinity” conformation.
- **Positive cooperativity:** *Increased affinity for O2 as the number of bound O2 molecules increase.*

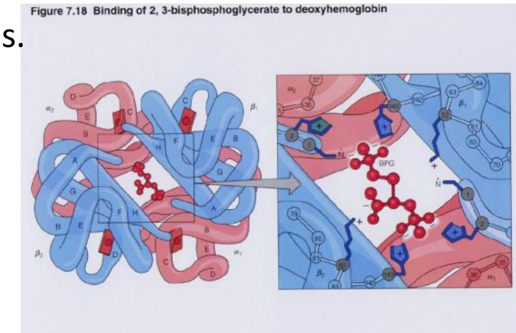


Factors affecting oxygen affinity

2, 3-bisphosphoglycerate (diphosphoglycerate) allows oxygen dissociation from hemoglobin (enhance the efficiency of unloading oxygen in tissues)

- 2, 3 diphosphoglycerate: biproduct of glycolysis. Stabilizes the deoxyhemoglobin form.
- In tissues, oxygen is released from oxygenated hemoglobin by simple diffusion. In return, 2,3-BPG stabilizes deoxyhemoglobin in tense (T) configuration.
- 2,3-BPG facilitates oxygen delivery to tissues.
- 2,3 BPG is increased in individuals living in high altitudes.

Figure 7.18 Binding of 2, 3-bisphosphoglycerate to deoxyhemoglobin

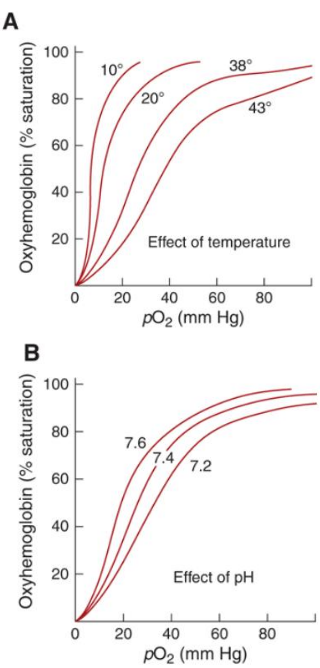


From Mathews and van Holde: *Biochemistry 2/e.* © The Benjamin/Cummings Publishing Co., Inc.

Temperature and pH

- Bohr effect demonstrates that the binding affinity of hemoglobin to oxygen is inversely related to acidity, tissue carbon dioxide concentration.

- Negative allosteric regulators (increased temperature, higher carbon dioxide and decreased pH) reduces oxygen-hemoglobin binding and enhance oxygen release to tissues.



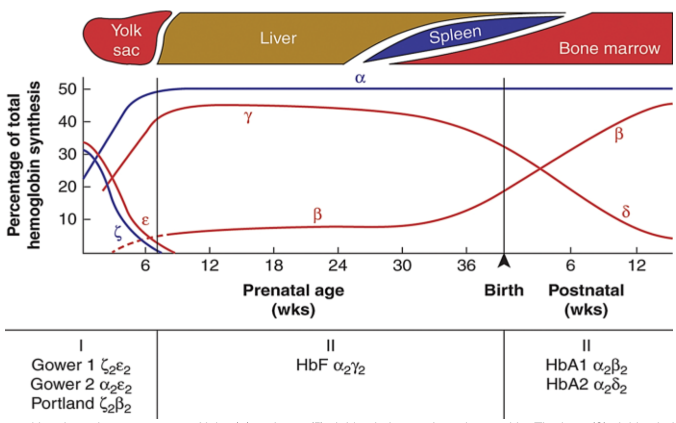
Types of hemoglobin

Adult hemoglobin:

- HbA: 2 α and 2 β polypeptide chains. Around 96% of total hemoglobin in adult RBC.
- Hemoglobin A1c (HbA1c): HbA and glucose. Glucose binds irreversibly to hemoglobin by glycation reaction(2-3 %).
- Hemoglobin A2 (HbA2): 2 α and 2 δ polypeptide chains. It is a normal variant of hemoglobin and represents 2-3% of total hemoglobin in adult RBC.

Fetal hemoglobin (HbF)

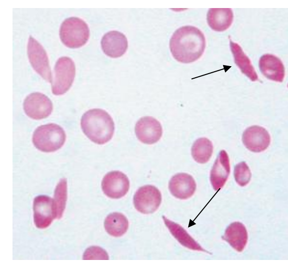
- 2 α and 2 γ polypeptide chains. Predominant during gestation, replaced by HbA by 6-12 months of age. Fetal hemoglobin has higher affinity to oxygen compared to adult. Facilitates/favors oxygen extraction in the the growing fetus

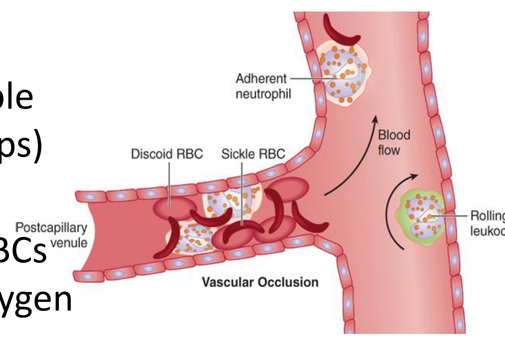


Hemoglobin types transition throughout pregnancy. Alpha (α) and zeta (ζ) globin chains are interchangeable. The beta (β) globin chain can be replaced by epsilon (ε), gamma (γ), or delta (δ) globin chains. In embryonic life (stage I), ζ and ε globin chain production is gradually replaced by α and γ globin chain production. Hemoglobin F is the main type in fetal life (stage II). After birth (stage III), HbA1 predominates.

Hemoglobin S "HbS"

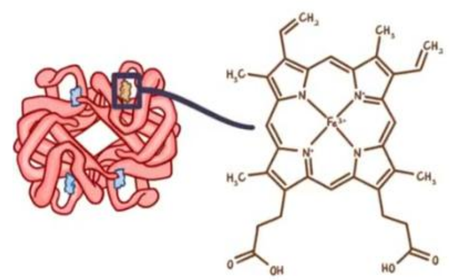
- HbS is a mutated form of HbA due to point mutation in *HBB gene* responsible for β-globin synthesis (switch from glutamic acid to valine amino acid).
- HbS is responsible for sickle cell disease (autosomal recessive disease). HbS is soluble when oxygenated but it polymerizes (clumps) when deoxygenated.
- RBCs may have altered shapes. Crescent RBCs structures can block the flow under low oxygen pressure.





Methemoglobin

- Methemoglobin = unfunctional hemoglobin
- Methemoglobin is an oxidized form of hemoglobin with iron in the FERRIC state (instead of ferrous state).
- Under physiological conditions, methemoglobin reductase keeps the methemoglobin level <1 % of our hemoglobin.



Myoglobin

- Myoglobin is a heme-containing globular protein normally found in the straited muscles.
- Myoglobin is formed from a single α-polypeptide chain bound to a heme molecule and one oxygen molecule.
- Myoglobin binds to oxygen more tightly than hemoglobin. Therefore, it can extract oxygen from blood. Mainly functions as an oxygen storage protein to support the high energy demands of contracting muscles.
- Elevated plasma myoglobin levels are indicative of muscle injury and hence myoglobin leakage to the circulation.

Heme & globin Disorders

Porphyria

Porphyria group of rare metabolic disorders that affects enzymes in the heme biosynthesis leading to the overproduction of intermediates like porphyrin or its precursor aminolaevulinic acid (ALA).

Seven subtypes of porphyria (hepatic and erythropoietic), both inherited and acquired.

Symptoms: acute and intense abdominal pain, neurologic symptoms including pain the extremities or chest, insomnia, anxiety or confusion or CVS symptoms (tachycardia and hypertension), photosensitivity, urine discoloration and in some cases erythrodontia or red-brownish teeth.

Severe case of porphyria showing extreme red discoloration of teeth.

Porphyria; skin blisters

Home - American Porphyria Foundation
porphyria picture | Medical Pictures Info - Health Definitions Photos

Dental manifestations of dermatologic conditions (researchgate.net)

Pathophysiology:

Common types: Acute intermittent porphyria and hepatic porphyria cutanea tarda

Acute attacks can be triggered by several drugs, infection, fasting, alcohol, steroid hormones.

Management:

Patient education and avoid triggers (alcohol, barbiturates and clonazepam and some NSAIDs).

Panhematin® therapy (IV) can be used to treat an acute attack. It acts as an enzyme inhibitor for heme biosynthesis enzymes.

Methemoglobinemia

Methemoglobinemia occurs due to

a) Decreased ability to reduce the oxidized hemoglobin (ferric state, genetic causes).
b) Exposure to chemicals or medications that increases the production of oxidized hemoglobin (acquired). Triggering medications include nitroglycerin, sulfamethoxazole, prilocaine or lidocaine/prilocaine combination.

Symptoms: Cyanosis, fatigue, weakness, headache, metabolic acidosis, seizures, coma, and death.

Management: Methylene blue can reduce the oxidized Hb (severe cases).

Carbon monoxide poisoning

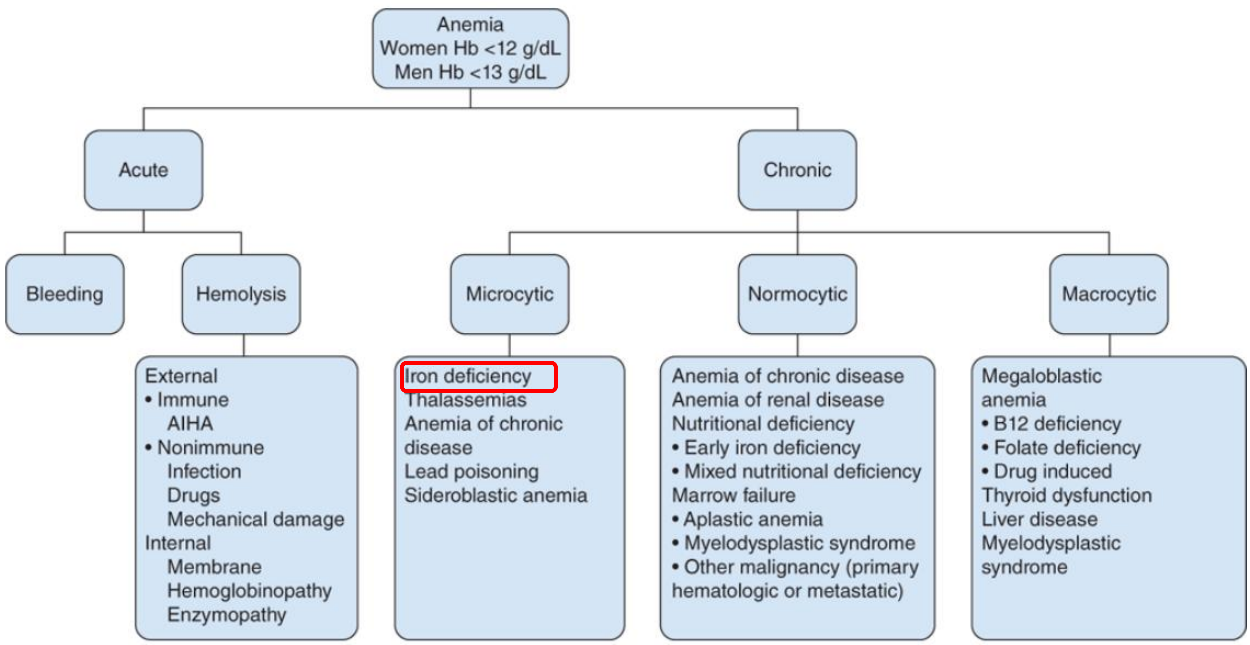
Carbon monoxide (CO) is a colorless, odorless, and nonirritating gas.

Produced by incomplete burning of carbon-containing materials like fireplace, smoke inhalation in fires, car exhaust fumes, etc.

Mechanism of toxicity: carbon monoxide has higher affinity for hemoglobin (250 times) than oxygen. Reduces oxyhemoglobin saturation and decreases blood oxygenation.

Symptoms: headache, nausea, myocardial or brain ischemic injuries. Carbon monoxide poisoning can be fatal.

Management: supplemental oxygen via high flow oxygen or hyperbaric chambers to prevent cognitive adverse events.



View Large | AccessMedicine | McGraw Hill Medical (cnsu.edu)

Physiological Adaptation to Anemia

Anemia is diagnosed if Hb <13 g/dL in men & <12 g/dL in female.

- Hb Level ≥ 10 may be asymptomatic.
- Moderate anemia: Hb <10 g/dL; Hb: 8-9 dyspnea and fatigue.
- Severe anemia: Hb <8 g/dL; Hb 6-7 can cause syncope; <5 can lead to HF.

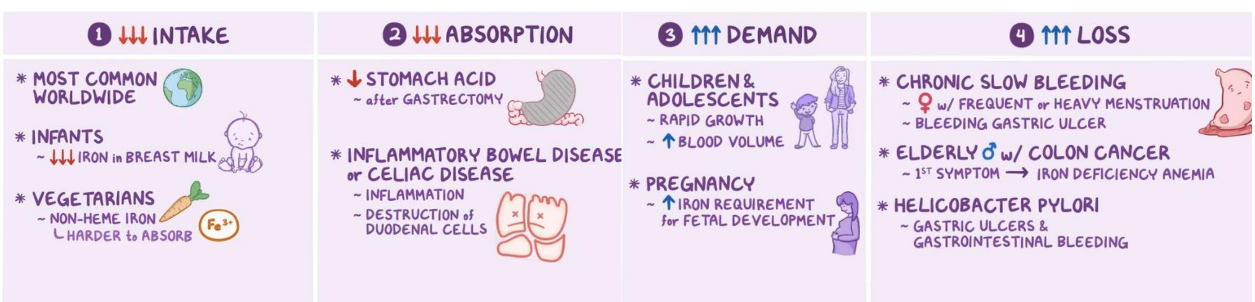
Physiological adaptation: increased erythropoietin production, enhanced oxygen extraction, increased cardiac output, and redistribution of blood flow to vital organs.

Preoperative anemia can increase postoperative morbidity and mortality. Hb <10 g/dl: corrective measures; Hb<8 g/dl "severe anemia" (refer/consider elective surgery).

Iron deficiency anemia

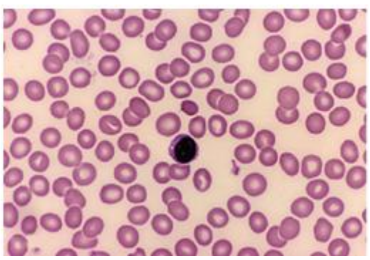
RBCs morphology	Hypochromic microcytic anemia
Prevalence	Common specially in children and females at childbearing age
Symptoms	fatigue, headache, exercise intolerance, pale skin, pale gum, brittle-spoon-shaped nails (koilonychia), pica
Diagnostic features	low RBCs count, mean corpuscular volume, serum ferritin and hemoglobin levels (females <12 gm/dl and males <13.5 g/dl)
Treatment	iron supplements

Causes

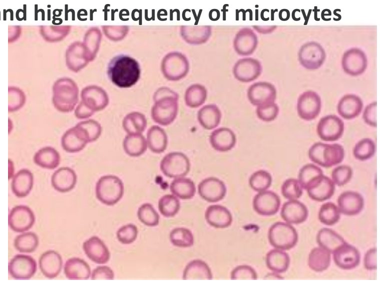


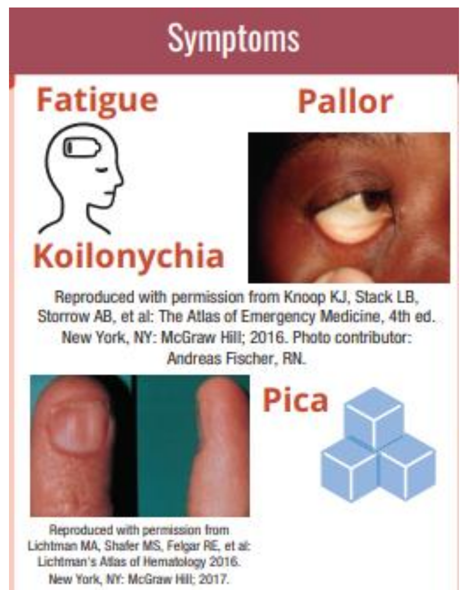
1 ↓↓↓ INTAKE
* MOST COMMON WORLDWIDE
* INFANTS
~ ↓↓↓ IRON in BREAST MILK
* VEGETARIANS
~ NON-HEME IRON
└ HARDER to ABSORB
Fe^{3+}
2 ↓↓↓ ABSORPTION
* ↓ STOMACH ACID
~ after GASTRECTOMY
* INFLAMMATORY BOWEL DISEASE or CELIAC DISEASE
~ INFLAMMATION
~ DESTRUCTION of DUODENAL CELLS
3 ↑↑↑ DEMAND
* CHILDREN & ADOLESCENTS
~ RAPID GROWTH
~ ↑ BLOOD VOLUME
* PREGNANCY
~ ↑ IRON REQUIREMENT for FETAL DEVELOPMENT
4 ↑↑↑ LOSS
* CHRONIC SLOW BLEEDING
~ ♀ w/ FREQUENT or HEAVY MENSTRUATION
~ BLEEDING GASTRIC ULCER
* ELDERLY ♂ w/ COLON CANCER
~ 1ST SYMPTOM ⟶ IRON DEFICIENCY ANEMIA
* HELICOBACTER PYLORI
~ GASTRIC ULCERS & GASTROINTESTINAL BLEEDING

Normal blood film. Normocytic-normochromic red cells with normal shape



Severe iron deficiency. hypochromic cells and higher frequency of microcytes





[Anemia, Iron Deficiency | The Infographic Guide to Medicine | AccessMedicine | McGraw Hill Medical (cnsu.edu)]

Platelets and Hemostasis

Platelets

Platelets (thrombocytes) are anucleated disk-shaped cell fragments that arise from megakaryocytes in bone marrow.
Life span is around 8-10 days.
Normal count is 300,000/400,000 -150,000 platelets/µl blood

- ≥ 100,000 = normal hemostasis.
- ≤50,000 increased bleeding risk.
- 20,000-10,000 spontaneous bleeding may occur.
- ≤ 10,000= life-threatening hemorrhage

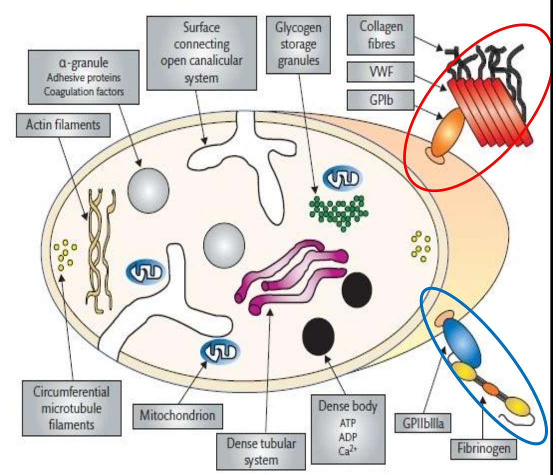
Platelet ultrastructure:

1- α-granules: contain adhesive proteins and coagulation factors in the cytoplasm.

2- Actin filaments & microtubules to maintain platelet shape, and facilitate shape change upon activation.

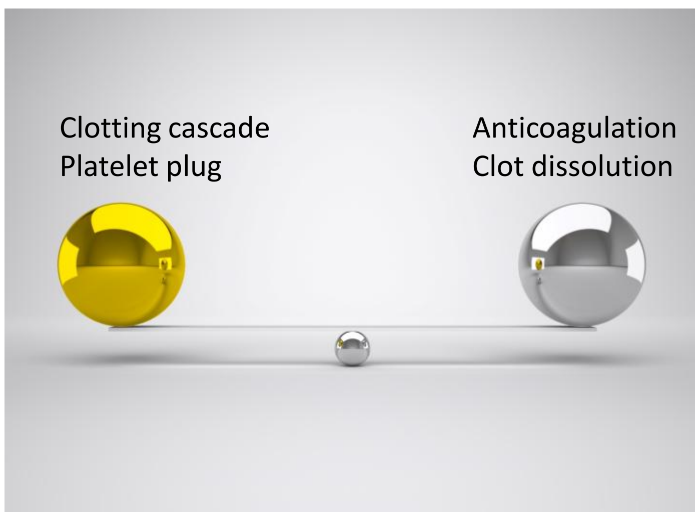
3- Platelet exterior surface is coated by a layer of glycoproteins and glycolipids known as the *glycocalyx.*

- GPIb receptor: binds to Von Willebrand factor "VWF" to facilitate platelet plug formation.
- GPIIb/IIIa receptor that binds to fibrinogen.



Adapted from Moore, Gary, et al. *Haematology*, (3rd Edition)

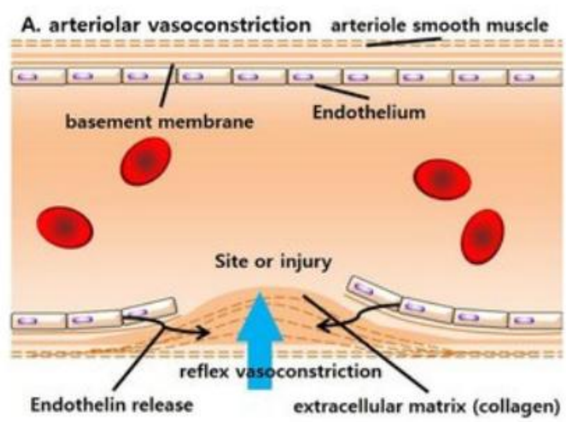
Hemostasis


Clotting cascade
Platelet plug
Anticoagulation
Clot dissolution

- Hemostasis can be considered as a host defense system that aims to maintain integrity of the high-pressure closed vascular system.
- Involves multiple steps:
  - Vasoconstriction
  - Primary hemostasis (platelet plug)
  - Secondary hemostasis (coagulation cascade).
  - Fibrin clot remodeling “fibrinolysis”.

A- Immediate Vasoconstriction

- Following vascular injury, immediate vasoconstriction is mediated by a reflex mechanism of the CNS.
- Damaged blood vessel constrict to limit blood loss.
- Endothelin, synthesized by endothelial cell lining, acts as a potent vasoconstrictor.



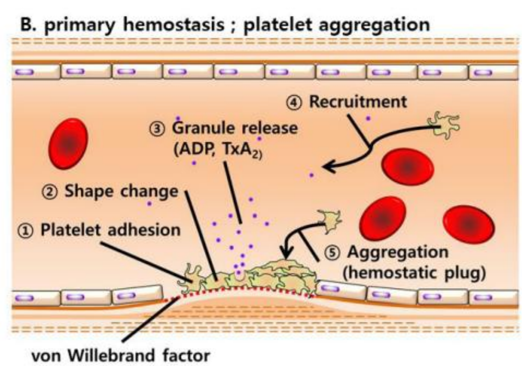
Era of Bloodless Surgery: Spotlights on Hemostasic Materials and Techniques (researchgate.net)

B- Primary hemostasis
"Platelet plug"

1- Damage to the endothelial lining of blood vessels exposes collagen on basement membrane and von Willebrand factor (vWF).

A) Collagen facilitates platelet adherence to the damage site via the glycoprotein coat of platelets.

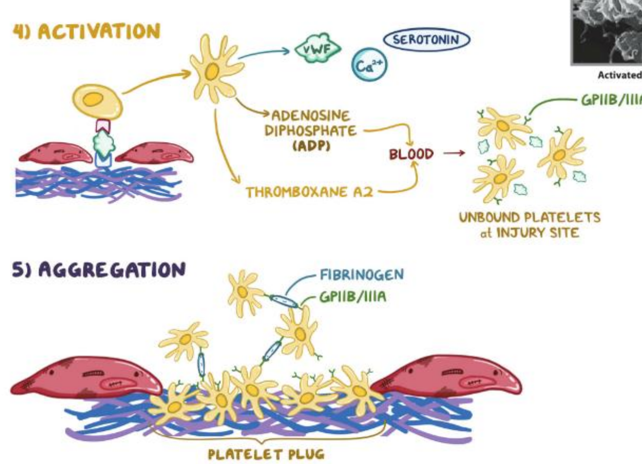
B) vWF acts as a bridge between collagen and the GP1b receptor of platelets to further enhance platelet adhesion.



Primary hemostasis (*cont.)*

2- Platelet activation: change shape, release granules causing the recruitment of more platelets.

3- Platelets aggregate together and form a hemostatic plug.



C- Secondary hemostasis
“Clotting cascade”

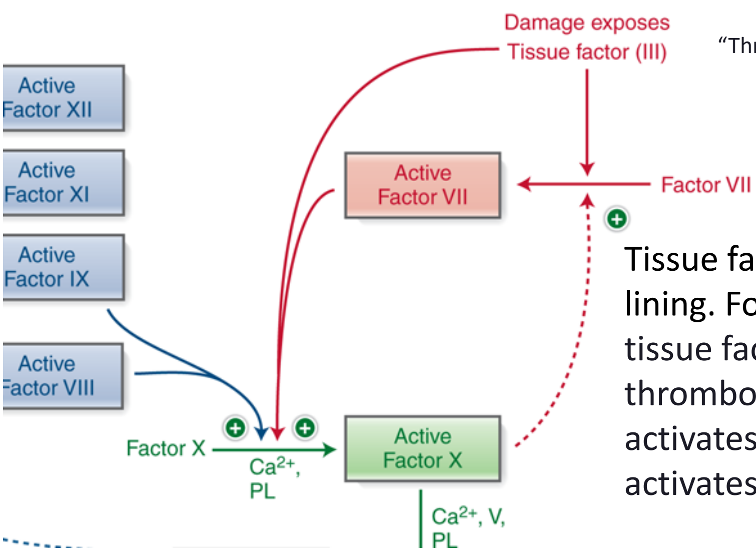
- Clotting cascade aims to amplify and propagate coagulation reaction.
- Includes 13 coagulation factors. Synthesized in liver. Soluble in plasma, as zymogens (I, II,…), activated form (Ia, IIa,….). Ends w insoluble fibrin clot.
- Composed of:

Intrinsic pathway	
Collagen activates factor XII → → → Xa	**Common pathway**
Extrinsic pathway	Factor Xa → Thrombin → Fibrin
Tissue factor activates Factor VII → Xa	

Clinical use of coagulation tests - UpToDate (cnsu.edu)

Extrinsic pathway

(tissue-factor dependent)



“Thromboplastin”

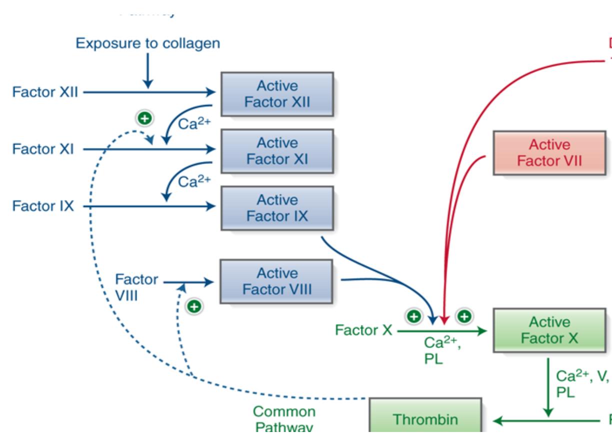
Tissue factor is present in endothelial lining. Following blood vessel damage, tissue factor (aka factor III or tissue thromboplastin) is released, and it activates factor VII which subsequently activates factor X and prothrombin

Intrinsic Pathway

Following exposure to collagen at the injury site, factor XII is activated.

Activation of factor XII leads to the sequential activation of factors XI, IX, VIII and factor X.

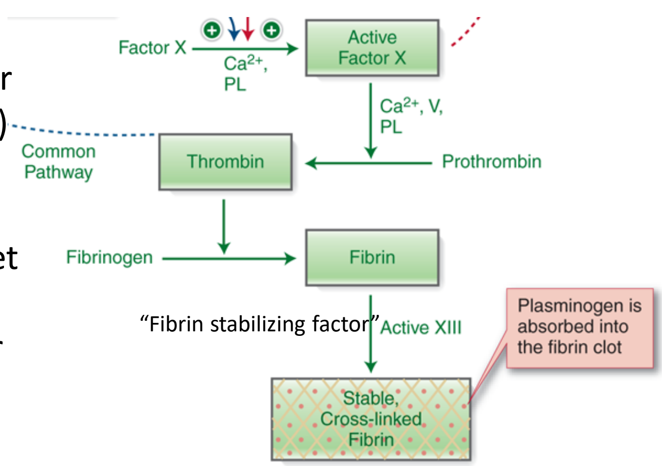
Calcium (factor IV) is a required co-factor for several intrinsic pathway steps.

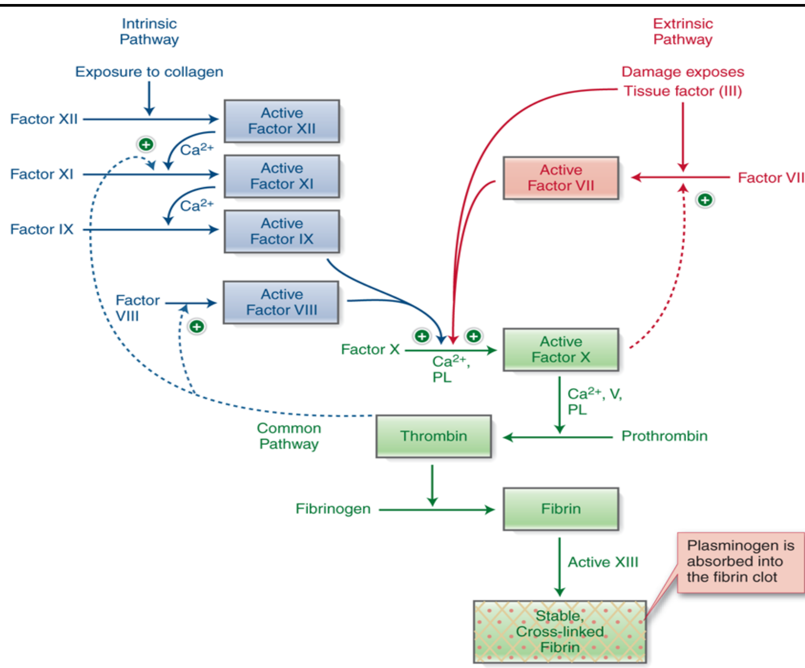


Thrombin can activate factors VIII or XI via a positive feedback mechanism. vWF factor stabilizes factors VIII. Therefore, low levels of vWF will affect the intrinsic pathway and platelet plug formation.

Common pathway

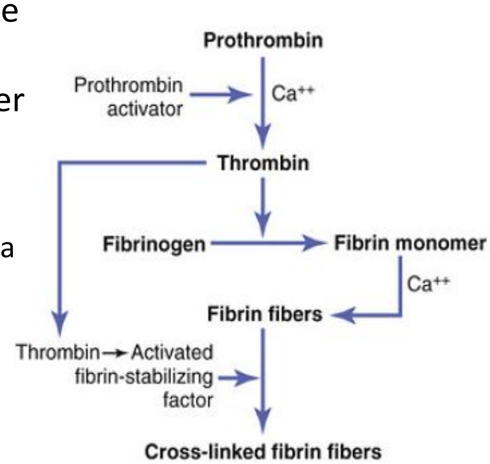
- In the presence of phospholipids and calcium (as cofactors), activated factor X converts prothrombin (aka, factor II) into thrombin (aka, factor IIa).
- Thrombin converts fibrinogen (aka, factor I) that was bound to the platelet plug to insoluble fibrin polymer.
- Fibrin polymer is crosslinked by factor XIII (aka, fibrin stabilizing factor) to form a hard clot.




Intrinsic Pathway
Extrinsic Pathway
Exposure to collagen
Damage exposes Tissue factor (III)
Factor XII
Active Factor XII
Ca^{2+}
Factor XI
Active Factor XI
Ca^{2+}
Active Factor VII
Factor VII
Factor IX
Active Factor IX
Factor VIII
Active Factor VIII
Factor X
Ca^{2+}, PL
Active Factor X
Ca^{2+}, V, PL
Common Pathway
Thrombin
Prothrombin
Fibrinogen
Fibrin
Active XIII
Plasminogen is absorbed into the fibrin clot
Stable, Cross-linked Fibrin

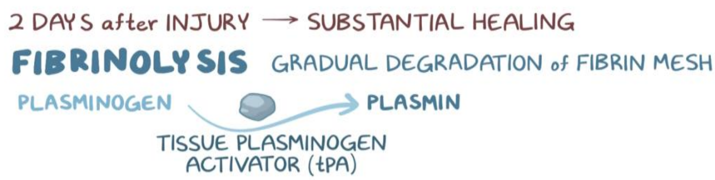
Prothrombin/Thrombin

- Prothrombin (factor II) is a plasma protein. Made by liver (requires vitamin K). Patients with liver disease or vitamin K deficiency might have higher risk of bleeding.
- Thrombin is an important clotting factor.
  - Can activate the intrinsic and common pathways via a positive feedback mechanism.
  - Activates the fibrin-stabilizing factor (factor XIII).
  - Can activate blood platelets.



VitalSource Bookshelf: Guyton and Hall Textbook of Medical Physiology E-Book

D- Fibrin clot remodeling



I) Clot dissolution

Plasminogen (aka profibrinolysin) is synthesized by liver and activated by tissue plasminogen activator (tPA) yielding plasmin. Plasmin causes fibrinolysis. Thrombin induces the secretion of tissue plasmin activator.

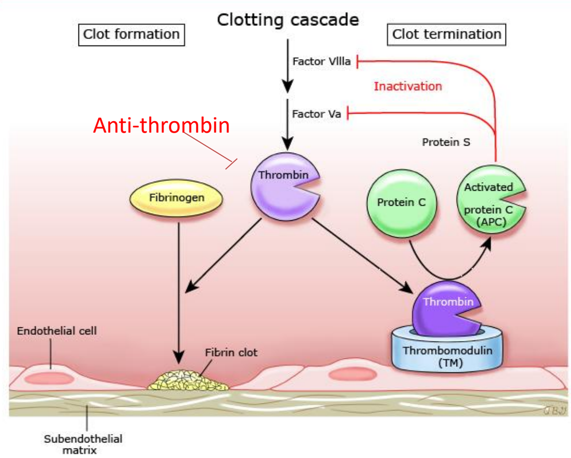
II) Clot remodeling.

Fibroblasts migrate into the wound, deposit extracellular matrix proteins and aid in wound remodeling. Tissue remodeling will also include dead cell phagocytosis by macrophages and vasculogenesis.

Regulation of coagulation

- Proteins C and protein S inactivate factors V and VIII of common & intrinsic pathway.
- Anti-thrombin or ATIII: Inactivates thrombin and factor X_a.
- To control the dynamics of clot dissolution, thrombin induces the release of tissue plasminogen activators from endothelial cells. Simultaneously, endothelial cells are secreting plasminogen activator inhibitor 1 and antiplasmin.

Hemostatic control mechanisms and termination of clotting



Refer to UpToDate for an overview of hemostasis regulation and the roles of specific procoagulant and anticoagulant factors in clinical thrombosis and hemostasis.

APC: activated protein C; TM: thrombomodulin.

Courtesy of Lawrence LK Leung, MD

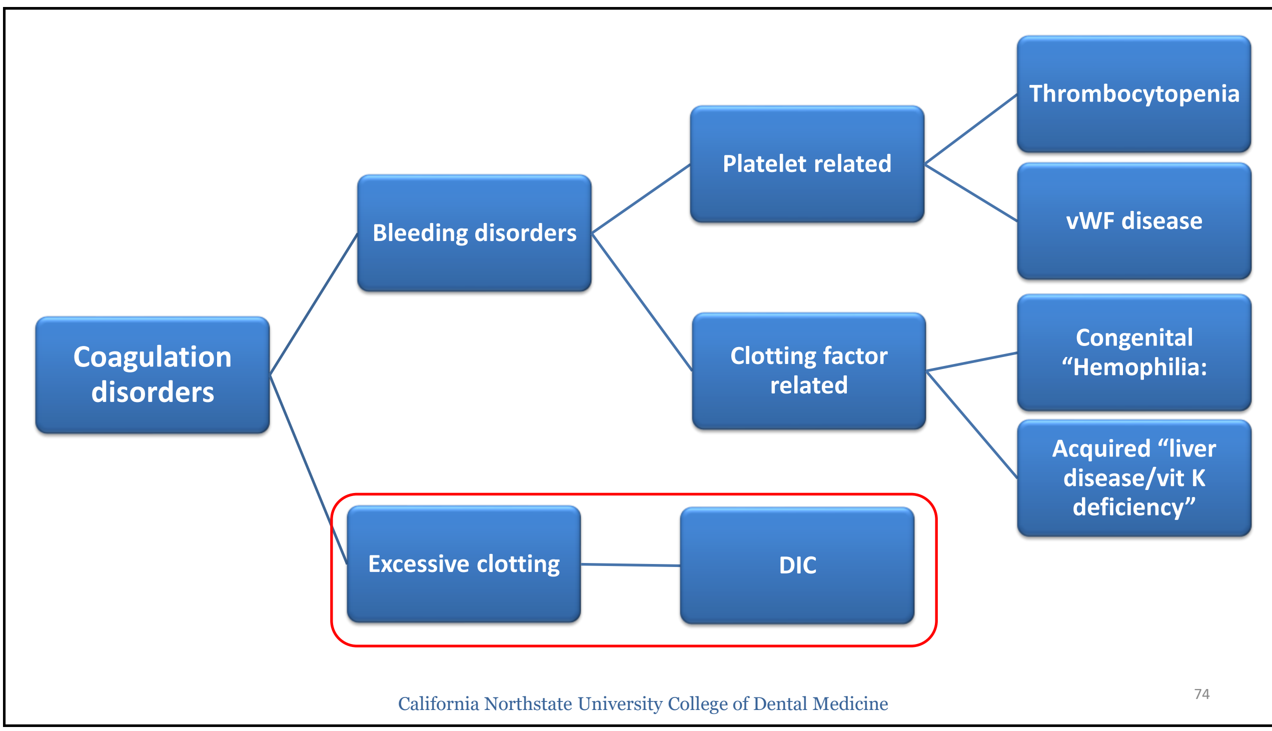
UpToDate

Role of vitamin K and Calcium

- Vitamin K is a fat-soluble vitamin abundant in green leafy vegetables.
- Vitamin K is stored in the liver, and it cycles between a reduced and oxidized forms.
- Reduced form of vitamin K acts as an essential cofactor for the activation of multiple coagulation factors as well as proteins C & S.
- Calcium: cofactor (factor IV) essential for activation of several cofactors in the intrinsic, extrinsic and common pathways. Released by activated platelets. Chelation of calcium by citrate ions can prevent coagulation.

Liver Diseases & Hemostasis

Etiology	Decreased production of clotting factors (factor VIII as well as vitamin k-dependent clotting factors). Reduced clearance of activated clotting factors in severe liver diseases. Decreased production of thrombopoietin.
Symptoms	Jaundice, increased risk for several bleeding disorders
Diagnostic features	prolonged PT and PTT. Elevated liver enzymes and INR value



Coagulation disorders
Bleeding disorders
Platelet related
Thrombocytopenia
vWF disease
Clotting factor related
Congenital “Hemophilia:
Acquired “liver disease/vit K deficiency”
Excessive clotting
DIC
California Northstate University College of Dental Medicine
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Disseminated intravascular coagulation (DIC)	
Causes	Multiple etiologies, usually caused by inflammation secondary to a severe infection, severe immune reaction, organ damage or cancer. Sepsis is the most common trigger of DIC
Pathophysiology	According to the disease stage. Tissue factor from injured vessels plays a central role in activating the blood coagulation pathway. Early phases is characterized by coagulopathy. Later stages, characterized by bleeding due to overconsumption of platelets and bleeding factors.
Symptoms	Blood in urine, chest pain, bleeding from nose, gum or mouth, dot-like or large patches of bruising in the skin
Diagnostic features	prolonged PT, low platelet count, low fibrinogen level
Treatment	Treatment of underlying cause and supportive blood transfusions

Disseminated intravascular coagulation: purpura fulminans Extensive cutaneous infarction with hemorrhage involving the hand. Similar lesions were on the face, the other hand, and the feet.

Source: K. Wolff, R.A. Johnson, A.P. Saavedra, E.K. Roh: Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology, Eighth Edition: www.accessmedicine.com

[SKIN SIGNS OF HEMATOLOGIC DISEASE | Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology, 8e | AccessMedicine | McGraw Hill Medical (cnsu.edu)]

Hematological Parameters

Parameter	Men	Women
Hemoglobin (g/dL)	13.6 to 16.9	11.9 to 14.8
Hematocrit (%)	40 to 50	35 to 43
RBC count (× 10^6/microL)	4.2 to 5.7	3.8 to 5.0
MCV (fL)	82.5 to 98	
MCHC	32.5 to 35.2	
RDW (%)	11.4 to 13.5	
Reticulocyte count (x 10^3/microL or x 10^9/L)	16 to 130	16 to 98
Platelet count (× 10^3/microL)	152 to 324	153 to 361
WBC count (× 10^3/microL)	3.8 to 10.4	

Adapted from Uptodate
https://www-uptodate-com.ezproxy.cnsu.edu/contents/diagnostic-approach-to-anemia-in-adults

Blood Count

Complete Blood count (CBC) counts the different blood cell populations. It also provide information regarding the size and hemoglobin content of the RBCs.

CBC with differential: "most common". Blood cell count with differential count for different white blood cell populations (lymphocytes, neutrophils, etc.).

CBC with differential and smear. A smear examines the blood cell morphology and color at the microscopic level.

Definitions

- Hematocrit value defines the percentage of red blood cells in the whole blood volume. Approximately 40% of the blood volume is occupied by blood cells. Hematocrit value reflects the blood viscosity.

Hematocrit value = red cell volume/total blood volume.

- Red cell distribution width (RDW) reflects the variation in RBC size.

- Mean corpuscular hemoglobin (MCH) is the average hemoglobin content per red blood cell. MCH= Hb/number of RBCs x 100 (amount/cell).

- Mean corpuscular hemoglobin concentration (MCHC) is the average hemoglobin content in the whole RBC population (Hb/HCT value=amount/volume)

- Mean corpuscular volume (MCV) is the average size and volume of the RBCs in the blood. MCV is instrumental in evaluated different types of anemia. Decreased in microcytic anemia (<80) and increased in macrocytic anemia (>100)
- Reticulocyte count reflects the ability of bone marrow to produce RBCs (Normal 1-2 %). Increased count reflects bone marrow response to anemia (4-5 %). Can differentiate causes of anemia (low is bone marrow suppression, high in hemolysis).

Diagnosis of coagulation disorders

Test	Normal level	Diagnostic for
Prothrombin time (PT)	11-15 second	Extrinsic pathway
International Normalized Ratio (INR)	1 healthy 2-3 patients on Warfarin	Extrinsic pathway
Activated Partial thromboplastin time (PTT)	25-35 second	Intrinsic pathway
Platelet count	>150,000/ul	Thrombocytopenia
Von Willebrand antigen test	50-200 IU/dL	vWF disease

Prothrombin time (PT) measures the time it takes blood plasma to clot when exposed to the tissue factor (thromboplastin). Assesses the extrinsic and common pathways of coagulation. Normal range for the PT varies according to the laboratory and experimental settings

International normalized ratio monitors the efficacy of anti-coagulation therapy or the bleeding risk.

$INR = [\text{Patient PT} \div \text{Control PT}]^{ISI}$.

INR values > 3.5 indicate risk of bleeding

Activated Partial thromboplastin time (aPTT) measures the time it takes plasma to clot when exposed to calcium that activates the coagulation factors. It assesses the intrinsic and common pathways of coagulation.

vWF antigen screening is important to exclude the hereditary bleeding disorder (von Willebrand disease). Those patients will have low vWF levels along with longer PTT time.

Let’s apply: Anemia

Compare the impact of microcytic and macrocytic anemia on the MCV and MCHC.

Recommended resources